

ECO1465 Final Project

Social Media, News Consumption, and Polarization: Estimating
Heterogeneous Effects Based on Prior Polarization

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1 Introduction

Political polarization has—in the past decade—become a defining feature of Western democracies, and social media platforms are widely viewed as both amplifiers and accelerators of this phenomenon. Initially celebrated for democratizing access to information, enabling bottom-up participation, and even being a key driver of movements like the Arab Spring, platforms such as Facebook, Twitter (now X), YouTube, and TikTok have increasingly been associated with digital echo chambers and misinformation. Early research on online polarization found ideological clustering on Facebook as early as the mid-2010s.^{1,2,3} The effect of media on political and affective attitudes is nothing new, of course; previous research have found strong effects of traditional media exposure in inducing extreme attitudes in Weimar and Nazi Germany.⁴ However, recent high-profile episodes such as the Cambridge Analytica scandal in 2018, the spread of conspiratorial content surrounding the validity of the 2020 American Election, the COVID-19 pandemic and vaccines, and the online organizing strategies of the modern alt-right have fueled public concern that political discourse is no longer mediated by traditional media, but through opaque, private algorithms and profit structures that value user attention—positive or negative—at the expense of the public forum. In this particular aspect, social media greatly differs from traditional media, where content is shaped by individual editors intentionally trying to segment the market,^{5,6} instead of the distributed flows of algorithms. Whether this dynamic merely reflects pre-existing partisan

1. Eytan Bakshy, Solomon Messing, and Lada A. Adamic, “Exposure to ideologically diverse news and opinion on Facebook” [in eng], *Science (American Association for the Advancement of Science)* (Washington) 348, no. 6239 (2015): 1130–1132, ISSN: 0036-8075.

2. SETH FLAXMAN, SHARAD GOEL, and JUSTIN M. RAO, “FILTER BUBBLES, ECHO CHAMBERS, AND ONLINE NEWS CONSUMPTION” [in eng], *Public opinion quarterly* (Oxford) 80 (2016): 298–320, ISSN: 0033-362X.

3. Pablo Barberá, “Birds of the Same Feather Tweet Together: Bayesian Ideal Point Estimation Using Twitter Data,” *Political Analysis* 23, no. 1 (2015): 76–91, <https://doi.org/10.1093/pan/mpu011>.

4. Maja Adena et al., “RADIO AND THE RISE OF THE NAZIS IN PREWAR GERMANY” [in eng], *The Quarterly journal of economics* (Oxford) 130, no. 4 (2015): 1885–1940, ISSN: 0033-5533.

5. Sendhil Mullainathan and Andrei Shleifer, “The Market for News” [in eng], *The American economic review* 95, no. 4 (2005): 1031–1053, ISSN: 0002-8282.

6. Matthew Gentzkow and Jesse M. Shapiro, “What Drives Media Slant? Evidence From U.S. Daily Newspapers” [in eng], *Econometrica* (Oxford, UK) 78, no. 1 (2010): 35–71, ISSN: 0012-9682.

divides or actively deepens them remains an open question, but the stakes are clear: online discourse is now inseparable from the collective public forum.

Unlike earlier forms of mass communication, social media blurs the line between interpersonal interaction, news consumption, and political campaigning. Users not only receive information but actively shape its circulation through likes, shares, and comments, often amplifying emotionally charged or divisive content. Because of this fact, there is evidence for an inherent bias within social media networks toward misinformation. Vosoughi et al. (2018) found that false news—especially false political news—spreads faster than true news on Twitter.⁷ Importantly, this is predominantly due to human resharing rather than bots.

A distinction should be made, however, between the existence of polarization and whether the content itself—news and information in particular—is the cause of observed polarization. Research on social media news in particular has found that most ideological variation in online news is within-outlet as opposed to between-outlet⁸ and that most individuals online do not excessively polarize their news consumption.⁹ Additionally, there is significant heterogeneity in polarization among demographic groups, particularly age, suggesting the presence of exogenous factors driving political polarization.¹⁰

Recent work in economics has begun addressing this question with field experiments. They have found that social media platforms simultaneously inform and polarize,¹¹ that backfiring effects potentially exist for counter-attitudinal content exposure,¹² and that the primary driver of

7. Soroush Vosoughi, Deb Roy, and Sinan Aral, “The spread of true and false news online” [in eng], *Science (American Association for the Advancement of Science)* (United States) 359, no. 6380 (2018): 1146–1151, issn: 0036-8075.

8. Luca Braghieri et al., “Article-Level Slant and Polarization of News Consumption on Social Media,” *Available at SSRN 4932600*, 2024,

9. Gregory Eady et al., “How Many People Live in Political Bubbles on Social Media? Evidence From Linked Survey and Twitter Data,” *Sage Open* 9, no. 1 (2019), <https://doi.org/10.1177/2158244019832705>.

10. Levi Boxell, Matthew Gentzkow, and Jesse M. Shapiro, “Greater Internet use is not associated with faster growth in political polarization among US demographic groups” [in eng], *Proceedings of the National Academy of Sciences - PNAS* (United States) 114, no. 40 (2017): 10612–10617, issn: 0027-8424.

11. Hunt Allcott et al., “The Welfare Effects of Social Media” [in eng], *The American economic review* 110, no. 3 (2020): 629–676, issn: 0002-8282.

12. Christopher A. Bail et al., “Exposure to opposing views on social media can increase political polarization,”

polarization may be individual user activity as opposed to platform algorithms.¹³

Given the uncertain and evolving nature of the field, understanding how and when social media contributes to polarization is essential for assessing the risks of digital politics and evaluating potential policy interventions. This paper contributes to that effort by extending the field experiment of Levy (2021)¹⁴ to estimate conditional treatment effects of exogenous exposure to counter-attitudinal news. This research effort extends the literature by applying novel techniques to the data gathered from a recent field experiment. The main results from the original paper were replicated, finding a statistically significant effect on affective political polarization from subscribing Facebook users to counter-attitudinal news in their feed. Investigations into tree-based models gave hints at a potential heterogeneous treatment effect among the treated since a number of baseline endogenous variables were the most significant determinants of affective polarization.

2 Data and Summary Statistics

The core dataset is the survey dataset from Levy (2021), comprising 37,494 observations (with the unit of observation being an individual) and 212 variables. The data was gathered in 2018 by the author in the form of two Facebook surveys: a baseline survey at the beginning of the experiment in February-March 2018 and an endline survey approximately two months later. The experimental sample consists of individuals who completed the endline survey (17,635 individuals). The primary treatment variable is coded as a pro/counter-attitudinal treatment based on whether the outlets assigned were aligned with or opposite to the participants' prior political leanings. Control variables included in the original regression and the analyses in previous reports include prior

Proceedings of the National Academy of Sciences 115, no. 37 (2018): 9216–9221, <https://doi.org/10.1073/pnas.1804840115>.

13. Andrew M. Guess et al., “How do social media feed algorithms affect attitudes and behavior in an election campaign?,” *Science* 381, no. 6656 (2023): 398–404, <https://doi.org/10.1126/science.abp9364>.

14. Ro'ee Levy, “Social Media, News Consumption, and Polarization: Evidence from a Field Experiment” [in eng], *The American economic review* 111, no. 3 (2021): 831–870, ISSN: 0002-8282.

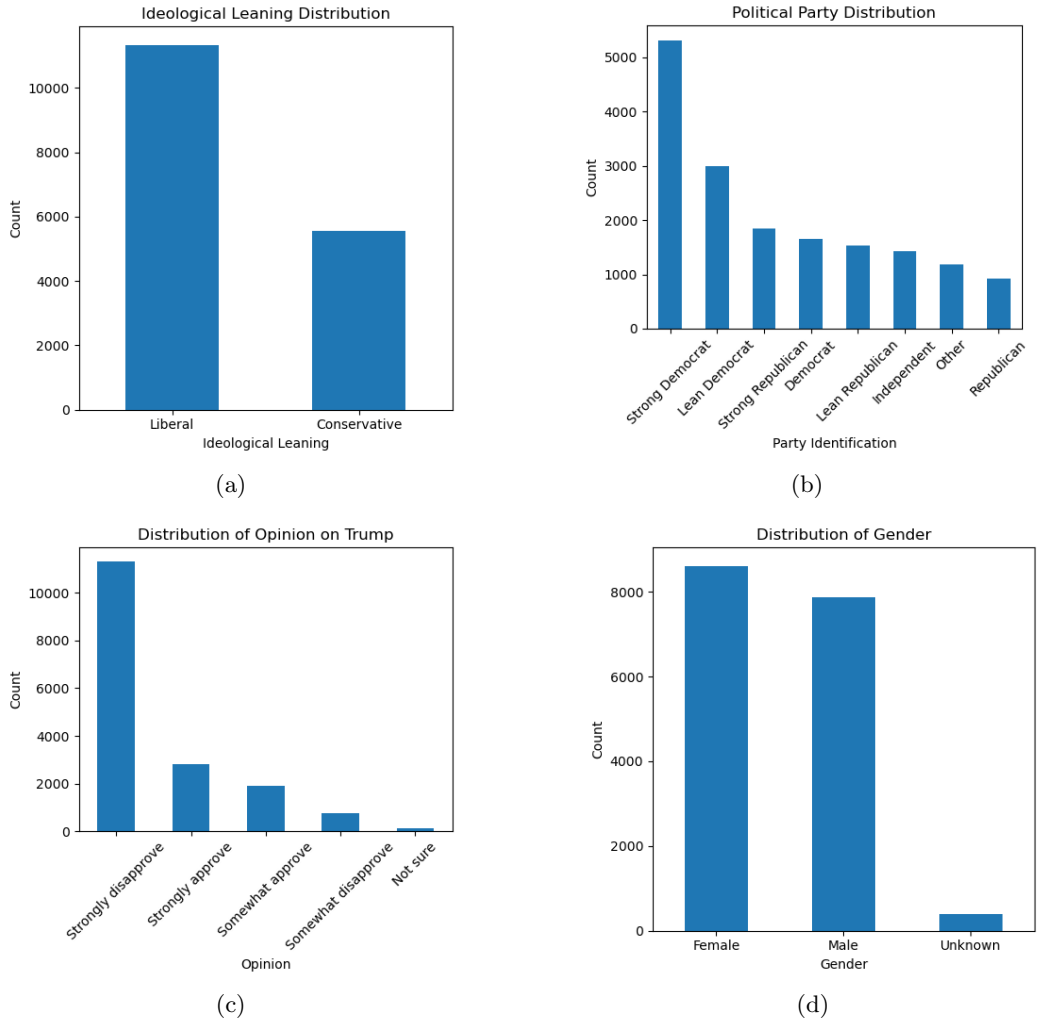
participant ideology (7-point scale between liberal and conservative), party identification (7-point scale between democrat and republican), Trump approval (5-point scale), gender, age, age squared, ideology leaning (binary between liberal and conservative), feeling thermometer difference between own and other party, and empathy difference between own and other party. After filtering out missing data, the final dataset comprises 16,896 observations. The author found no meaningful difference in attrition rates or observable characteristics between treatment groups. Given the large number of variables, the analysis will primarily focus on the outcome, treatment, and control variables the authors used for the primary regression. Since many of the variables are categorical, dummy variables were created for each category minus one to serve as model inputs.

Table 1 shows basic summary statistics for numerical data, and Figure 1 shows basic summary plots for categorical data. We can see from Table 1 that the average participant is approximately 47 years old, and the data capture a wide variation in age (note that the minimum age of 0 indicates a missing age value). We can also see that both the feelings thermometer difference and empathy difficult difference (defined as feeling thermometer toward own party minus feeling thermometer toward other party and finding it easy to see things from the participant’s own point of view minus the opposing party, respectively) have a positive mean, indicating that participants have existing affective biases towards individuals of their own political persuasion. Finally, we can see that the normalized Affective Polarization index has a significant rightward skew, indicating that on the extreme ends of the spectrum, individuals tend to be less polarized than those who are more polarized (higher values of the index indicate greater polarization). We can see from the distribution of categorical variables from the baseline survey (ideological leaning, party identification, and Trump approval) that the sample is skewed towards Liberals/Democrats and that the sample is slightly skewed towards women than men.

Table 1: Summary Statistics of Survey Numerical Variables

	Age	Feeling Thermometer Difference	Empathy Difficult Difference	Affective Polarization Index
mean	47.3	48.4	1.9	0.0
std	19.5	29.8	1.6	1.0
min	0.0	-100.0	-4.0	-5.6
25%	31.0	30.0	1.0	-0.7
50%	51.0	51.0	2.0	0.0
75%	64.0	70.0	3.0	0.7
max	89.0	100.0	4.0	2.3

Figure 1: Summary Plots of Categorical Variables



3 Regression and Model Results

Given that the paper employs an RCT design, there are minimal assumptions needed for identification. The balance table of Counter-attitudinal treatment and Pro-attitudinal treatment is displayed in Table 2. Differences in all variables across treatment arms are insignificant at standard significance levels.

Table 2: Balance table of variables by treatment (excluding Control)

	Counter	Pro	Difference	p-value
Age	46.254	46.379	-0.125	0.612
Feelings Thermometer Difference	47.882	48.080	-0.198	0.610
Empathy Difficult Difference	1.841	1.834	0.007	0.730
Took Endline Survey	0.461	0.463	-0.002	0.689
Ideology: Extremely liberal	0.159	0.162	-0.003	0.586
Ideology: Liberal	0.297	0.294	0.003	0.629
Ideology: Slightly liberal	0.133	0.133	-0.001	0.858
Ideology: Moderate; middle of the road	0.089	0.091	-0.002	0.648
Ideology: Slightly conservative	0.097	0.097	-0.000	0.956
Ideology: Conservative	0.164	0.164	0.000	0.962
Ideology: Extremely conservative	0.051	0.050	0.001	0.799
Ideology: Haven't thought much about this	0.010	0.009	0.001	0.246
Party Identification: Democrat	0.096	0.094	0.002	0.531
Party Identification: Independent	0.086	0.089	-0.002	0.506
Party Identification: Lean Democrat	0.175	0.181	-0.006	0.187
Party Identification: Lean Republican	0.091	0.094	-0.004	0.347
Party Identification: Other	0.074	0.074	-0.000	0.943
Party Identification: Republican	0.059	0.058	0.001	0.752
Party Identification: Strong Democrat	0.299	0.296	0.003	0.610
Party Identification: Strong Republican	0.121	0.115	0.006	0.125
Trump Opinion: Not sure	0.009	0.009	-0.001	0.673
Trump Opinion: Somewhat approve	0.120	0.119	0.001	0.829
Trump Opinion: Somewhat disapprove	0.049	0.049	-0.000	0.994
Trump Opinion: Strongly approve	0.175	0.174	0.000	0.967
Trump Opinion: Strongly disapprove	0.648	0.648	-0.001	0.927
Gender: Unknown	0.024	0.025	-0.001	0.659
Gender: Female	0.508	0.510	-0.003	0.669
Gender: Male	0.468	0.465	0.004	0.572
Ideo Leaning: Conservative	0.348	0.348	0.000	0.957
Ideo Leaning: Liberal	0.652	0.652	-0.000	0.957
N	12126	12100		

A similar table testing for differential attrition is found in Table 3. Note that this is grouped based on an included variable in the dataset and not based on missing values, hence the differences in N compared with later OLS regressions. In contrast to the balance table for the treatment arms, many of the differences in survey attrition are statistically significant (due to the large N), but are likely to be materially insignificant due to their small magnitude.

Table 3: Balance table of variables by attrition

	Both Surveys	Baseline Only	Difference	p-value
Age	47.072	45.283	1.789	0.000
Feelings Thermometer Difference	46.945	46.282	0.663	0.038
Empathy Difficult Difference	1.840	1.739	0.101	0.000
Ideology: Extremely liberal	0.172	0.141	0.031	0.000
Ideology: Liberal	0.297	0.276	0.021	0.000
Ideology: Slightly liberal	0.128	0.128	-0.000	0.929
Ideology: Moderate; middle of the road	0.103	0.124	-0.021	0.000
Ideology: Slightly conservative	0.088	0.100	-0.012	0.000
Ideology: Conservative	0.152	0.166	-0.014	0.000
Ideology: Extremely conservative	0.049	0.049	-0.000	0.932
Ideology: Haven't thought much about this	0.010	0.015	-0.005	0.000
Party Identification: Democrat	0.096	0.089	0.007	0.029
Party Identification: Independent	0.101	0.107	-0.006	0.080
Party Identification: Lean Democrat	0.173	0.169	0.004	0.366
Party Identification: Lean Republican	0.088	0.093	-0.005	0.085
Party Identification: Other	0.078	0.087	-0.010	0.001
Party Identification: Republican	0.054	0.055	-0.001	0.598
Party Identification: Strong Democrat	0.306	0.276	0.030	0.000
Party Identification: Strong Republican	0.105	0.124	-0.018	0.000
Trump Opinion: Not sure	0.008	0.013	-0.005	0.000
Trump Opinion: Somewhat approve	0.113	0.130	-0.016	0.000
Trump Opinion: Somewhat disapprove	0.052	0.059	-0.007	0.004
Trump Opinion: Strongly approve	0.161	0.177	-0.016	0.000
Trump Opinion: Strongly disapprove	0.666	0.622	0.045	0.000
Gender: Unknowwn	0.024	0.024	-0.000	0.896
Gender: Female	0.507	0.503	0.004	0.489
Gender: Male	0.469	0.472	-0.003	0.514
Ideo Leaning: Conservative	0.319	0.353	-0.034	0.000
Ideo Leaning: Liberal	0.652	0.614	0.039	0.000
N	17635	19859		

The OLS results excluding and including controls can be found below in Table 2. The OLS

model specification follows the form below where X_i is a vector of covariates with the effects of interest being β_1 , β_2 , and $\beta_1 - \beta_2$. Included interaction terms are empathy difference and feelings thermometer difference, interacted with the treatment to create a baseline indicator of heterogeneous treatment effects by measures of prior affective polarization.

$$Y_i = \alpha + \beta_1 Counter + \beta_2 Pro + X^\top \gamma + \epsilon_i \quad (1)$$

We can see that the results are consistent with those of the author, finding about a 0.03 unit gap between the counter-attitudinal treatment arm and the pro-attitudinal treatment arm in terms of affective polarization, with a statistically significant effect for the counter-attitudinal treatment and no statistically significant effect for the pro-attitudinal treatment. The addition of control variables (baseline ideology, party identification, trump favourability, gender, age, age squared, feelings thermometer difference, and empathy difficulty difference) greatly increases the R^2 value and decreases the effect size of the counter-attitudinal treatment, though it remains significant. When interaction terms are included, all treatment coefficients become insignificant but the coefficients on the interaction terms themselves are not significant. This indicates that, if there does exist a heterogeneous treatment effect, it is not captured well by a simple linear model.

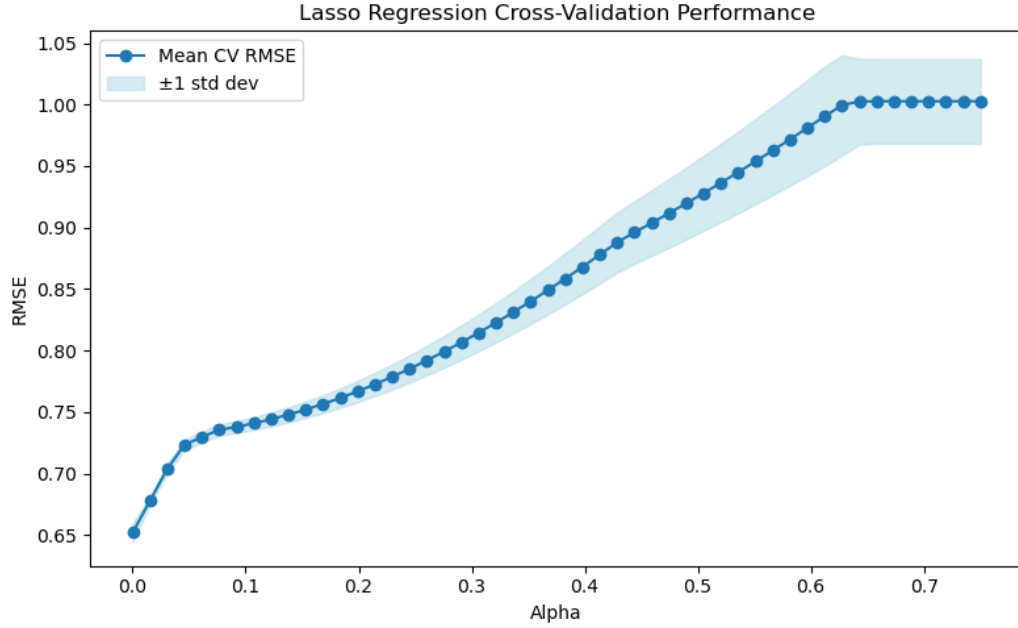
The application of LASSO and Ridge regression yielded insignificant results. There were no non-zero values of the regularization hyperparameter for either model that yielded better predictive results than the baseline OLS regression with controls. Cross-validation results are plotted in Figure 2 and parameter regularization paths are plotted in Figure 3.

Table 4: OLS Regression Results

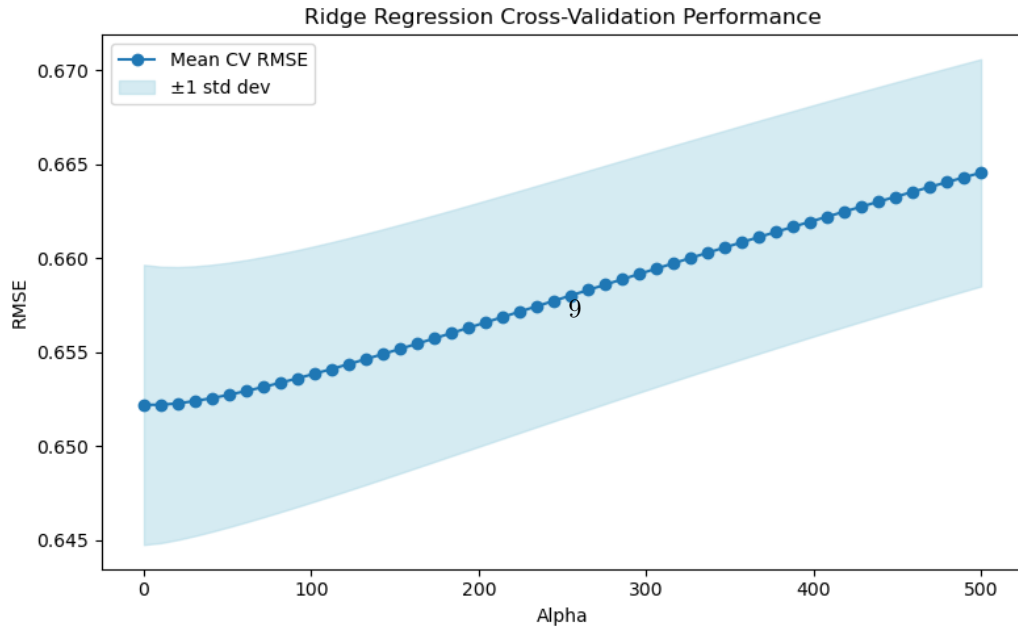
	Baseline	Controls Included	Controls and Interaction
Intercept	0 (0.013)	0.29*** (0.092)	0.28*** (0.093)
Counter-Attitudinal	-0.055*** (0.019)	-0.028** (0.012)	-0.027 (0.024)
Pro-Attitudinal	-0.022 (0.019)	0.005 (0.012)	0.029 (0.024)
Controls	No	Yes	Yes
Interaction	No	No	Yes
N	16896	16896	16896
R^2	0.001	0.579	0.579

Standard errors in parentheses
 * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 2: Lasso and Ridge CV results

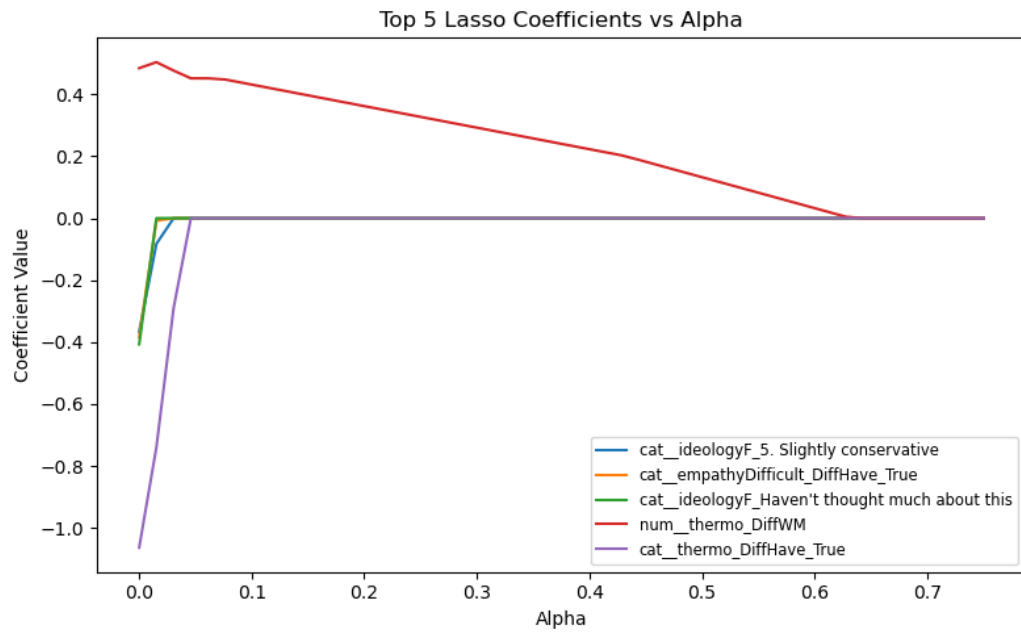


(a)

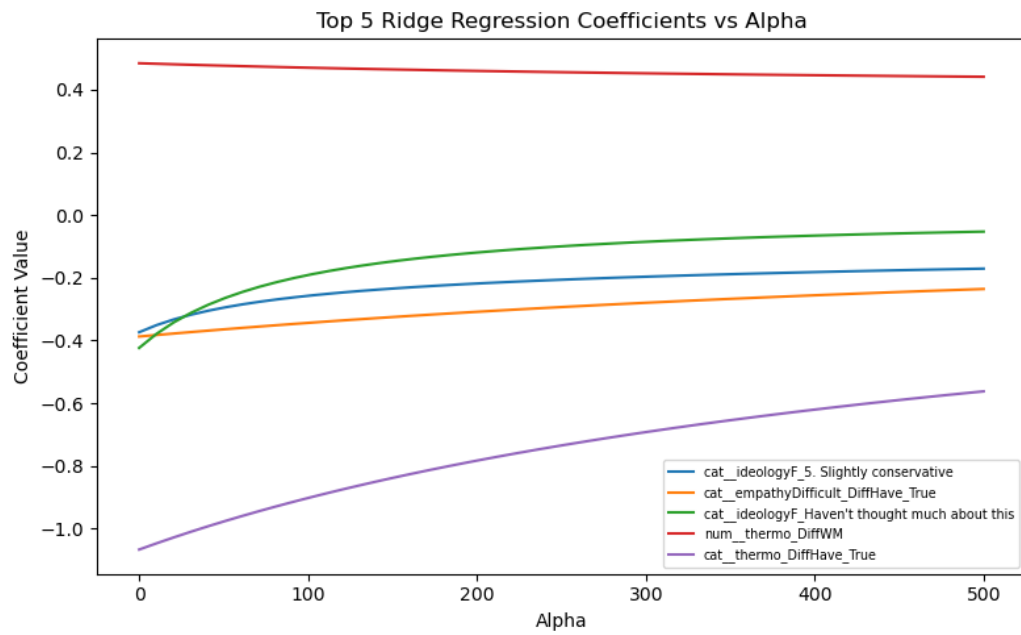


(b)

Figure 3: Lasso and Ridge Coefficient values



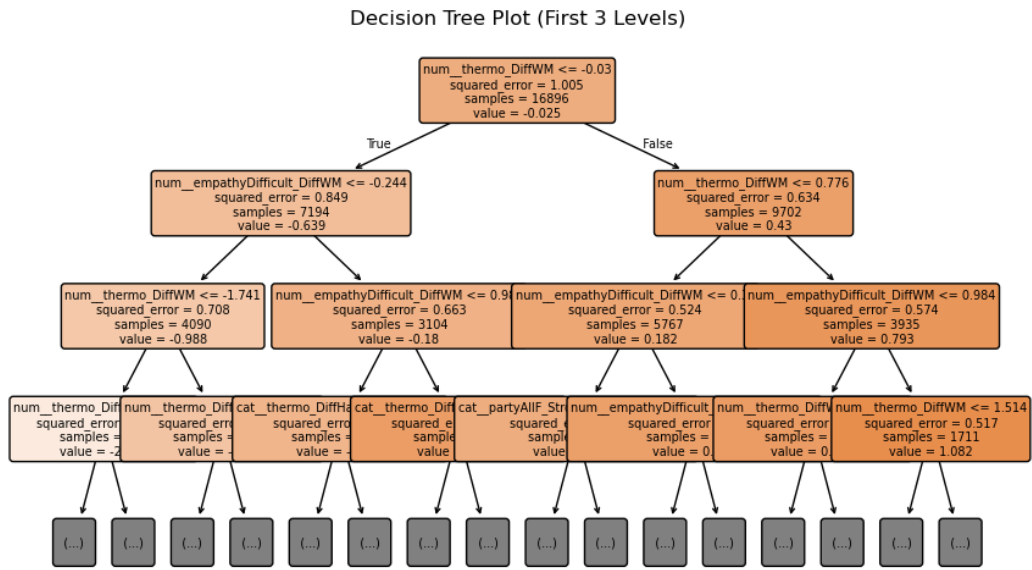
(a)



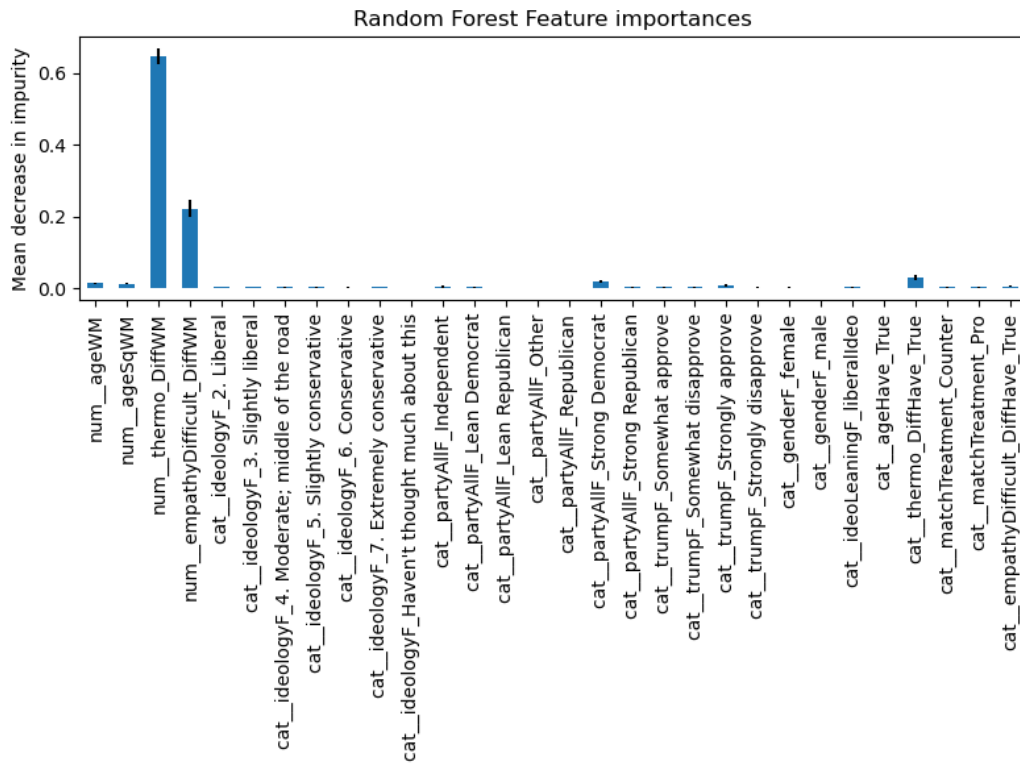
(b)

Tree models give us our first look at potential differences in treatment effects (Figure 4). Both the individual tree plot and the feature importance plot for the random forest model indicate that the treatment is quite insignificant for predicting affective polarization. Instead, the differences in own-party vs other-party feeling thermometer scores and empathy difficulties are the main determinants of affective polarization. This is in line with the results from OLS where the R^2 value increased by multiple orders of magnitude once covariates were included in the model.

Figure 4: Tree Plot and Random Forest Feature Importance



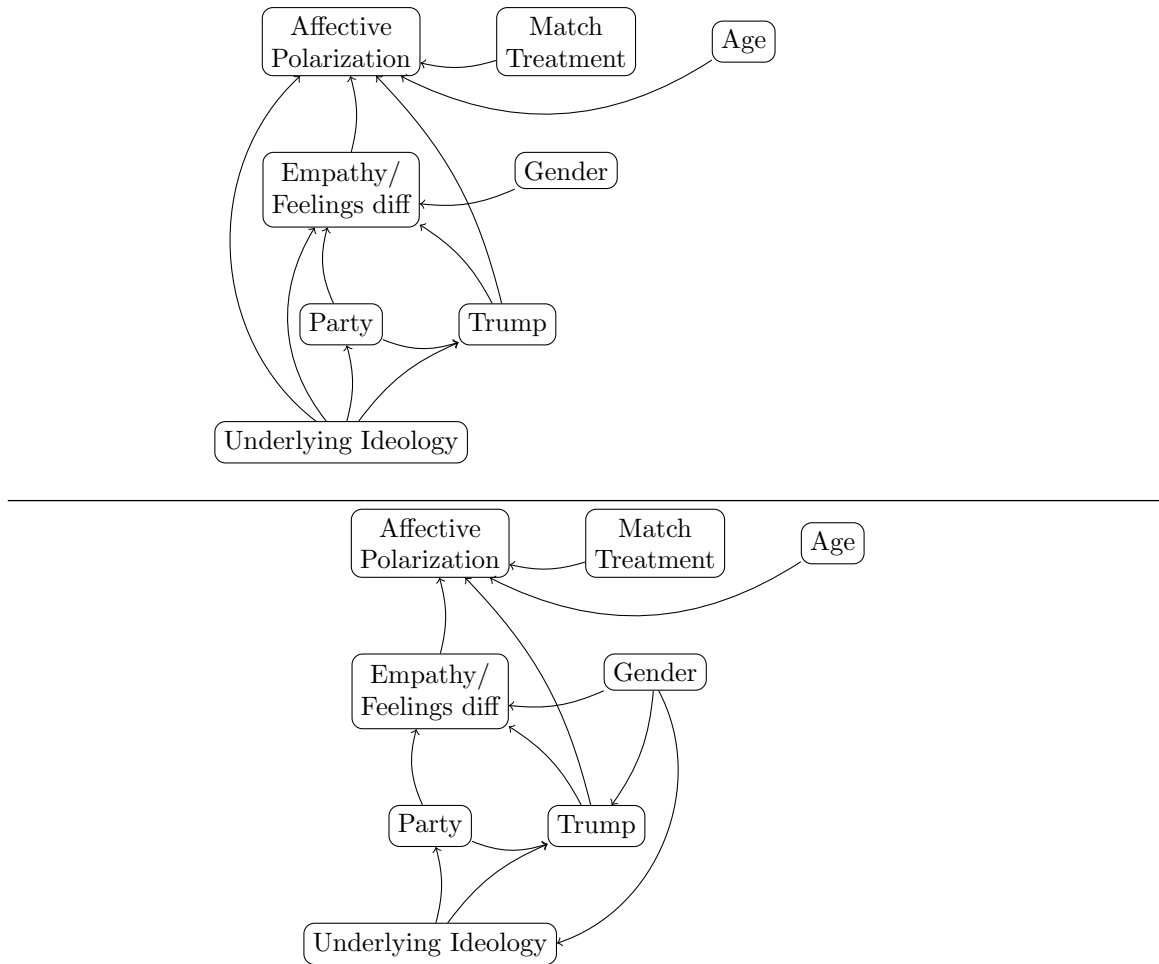
(a)



(b)

Regarding a causal DAG model, it is not particularly relevant in this case due to the RCT design. Additionally, due to the ill-defined nature of political and ideological concepts, identifying the exact DAG model is extremely difficult and subject to a multitude of conceptual criticisms. Nevertheless, in the interest of completeness I have proposed a few DAG models below, however I do not believe that any of them are either relevant to the analysis at hand or captures the true causal relation among the variables (observed or unobserved).

Figure 5: Two Possible DAG Models



When estimating propensity scores, since the original data is drawn from an RCT, we would expect any lack of overlap to be due to attrition as opposed to poor randomization. Indeed, when we estimate a propensity score for the three treatment classes (Control, Pro-attitudinal, Counter-attitudinal) using a logit model, we see good overlap between the two treatment groups but a slight mismatch between the treatment and control groups. However, when we estimate a propensity model using a flexible boosting model (LightGBM), all three groups have nearly identical overlap. This is strong evidence that matching is not an issue, especially when the main effect of interest (difference between pro and counter treatment groups) has no positivity issues even when using a simple linear model. Refer to Figure 6 below.

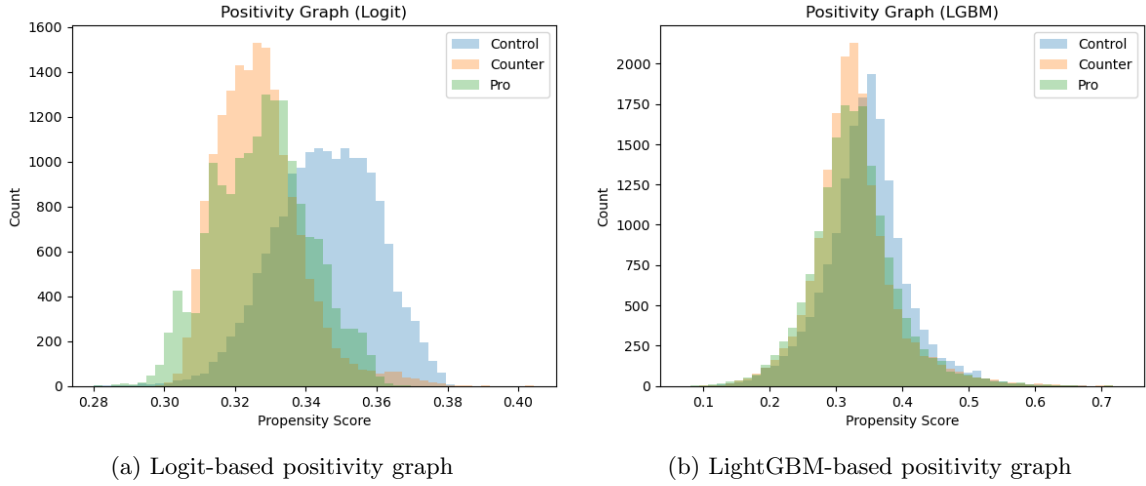


Figure 6: Positivity estimations

When meta-learners are applied, the predicted average treatment effects (ATE) of the difference between the pro and counter-attitudinal treatments are very similar to those of the author's original OLS model with all 3 estimated ATEs being within the 95% confidence interval of the OLS estimates. All 3 models were built with the LightGBMRegressor.

	T-learner	S-learner	X-learner
Average Treatment Effect	-0.029	-0.024	-0.029

Table 5: Average Treatment Effects for Different meta-learners

When the same effect is estimated with Doubly Robust ML (DRML), the effect sizes are very similar. As a robustness check, three underlying models were used to estimate the effect size: OLS, Random Forest, and Light Gradient Boosting. The propensity model was estimated with Light Gradient Boosting as well. Estimates for the ATE are shown in the table below. These results are all within the 95% confidence interval of the original OLS estimate as well as within the bootstrapped 95% confidence intervals of each other.

	OLS	RF	LGBM
Average Treatment Effect	-0.031	-0.027	-0.030

Table 6: DRML Average Treatment Effects

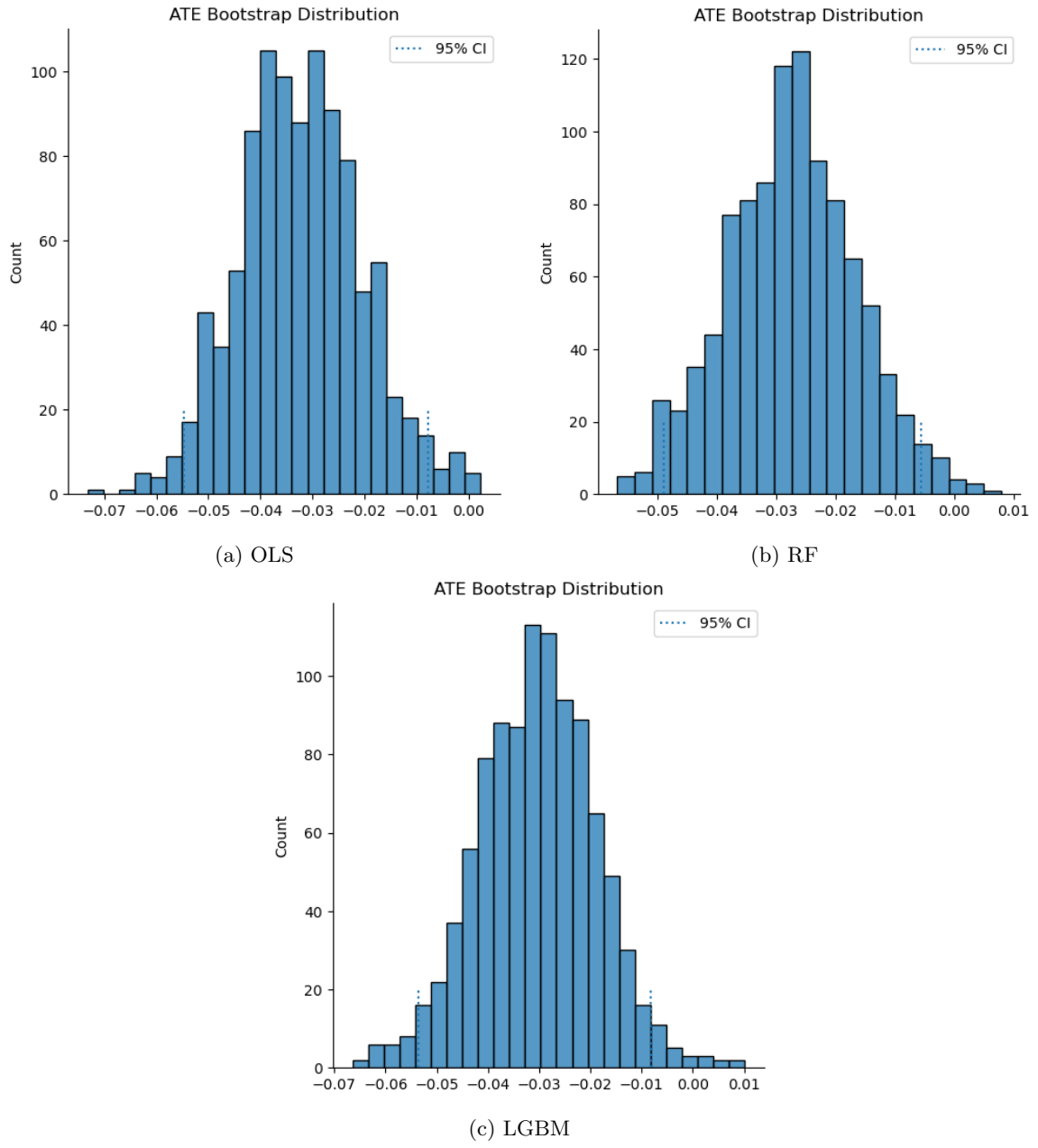


Figure 7: DRML 95% CI estimates of the ATE

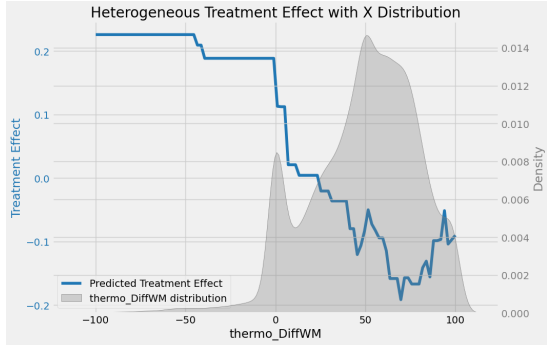
Similarly, when Double Machine Learning and Causal Forest models are applied, the results are similar to the original OLS estimates.

	Linear DML	Non-Parametric	GRF
Average Treatment Effect	-0.028	-0.029	-0.028

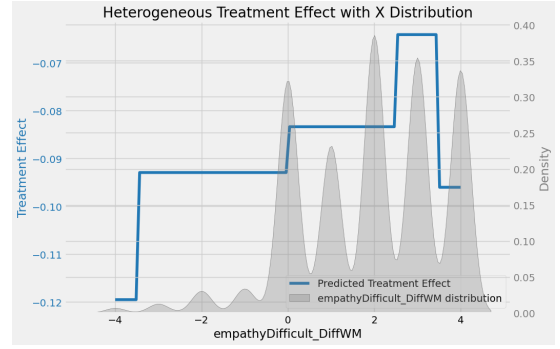
Table 7: DML and Causal Forest Average Treatment Effects

However, one of the main benefits of causal ML models is the estimation of Conditional Treatment Effects as opposed to simply the average treatment effects. Using the results from the linear regularized models and tree models, we can see that the feelings thermometer difference and empathy difference have the largest effects on the outcome variable. This is sensible as these variables represent prior polarization pre-treatment and there is likely a strong relationship between prior polarization and post-treatment polarization, given the small effect size of the treatment. All CATEs were estimated with simulated data with the value of all variables except the variable of interest set to their sample means and the variable of interest interpolated between its minimum and maximum values.

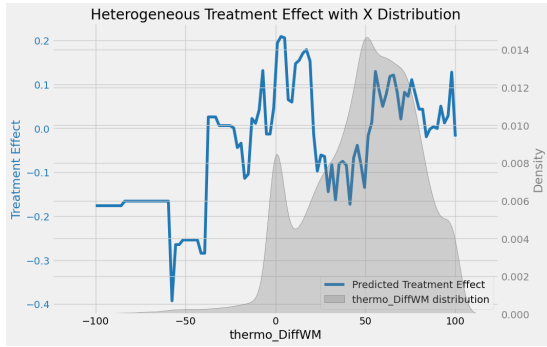
As can be seen in figures 8 and 9, the estimates for the CATE across both prior measures of polarization are variable and inconsistent between models. This is generally difficult to interpret and indicates the complexity of the underlying problem and of political economy problems in general. Especially since both variables (empathy difference and thermometer difference) are different but related measures of underlying polarization but they only have a medium correlation of about 0.44.



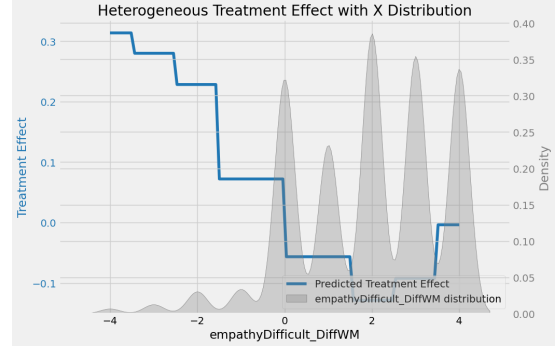
(a) S-learner Thermometer Difference CATE



(b) S-learner Empathy Difference CATE

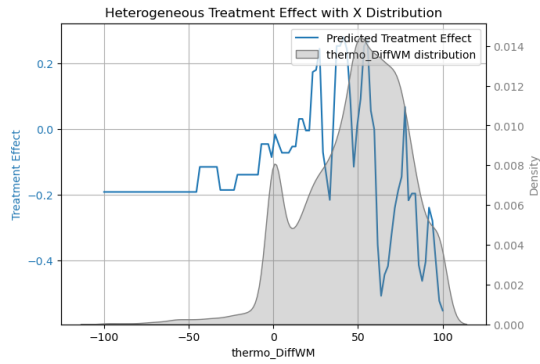


(c) X-learner Thermometer Difference CATE

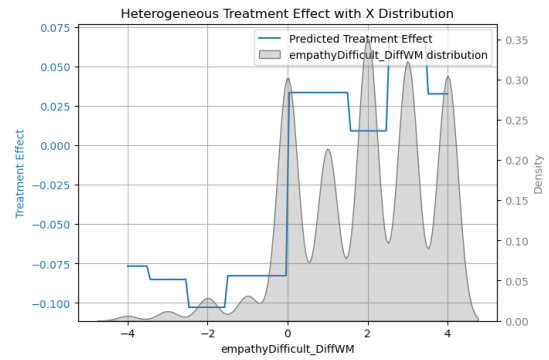


(d) X-learner Empathy Difference CATE

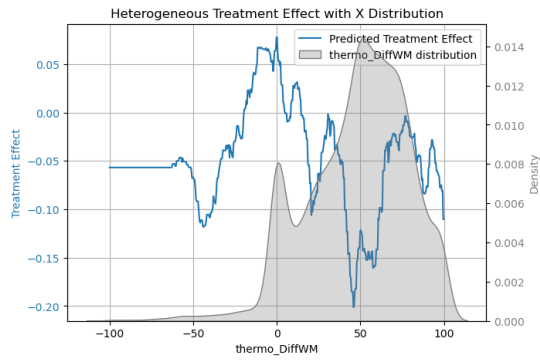
Figure 8: Meta-learner CATE estimates



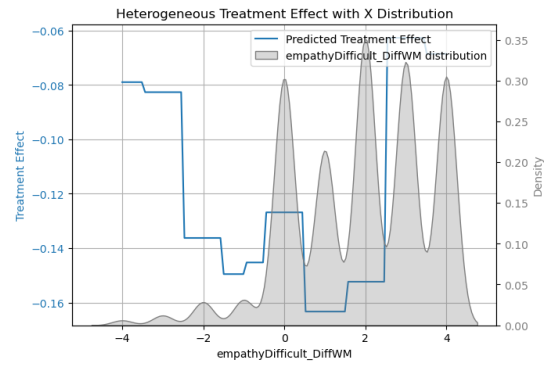
(a) DML Thermometer Difference CATE



(b) DML Empathy Difference CATE



(c) GRF Thermometer Difference CATE



(d) GRF Empathy Difference CATE

Figure 9: DML and GRF CATE estimates

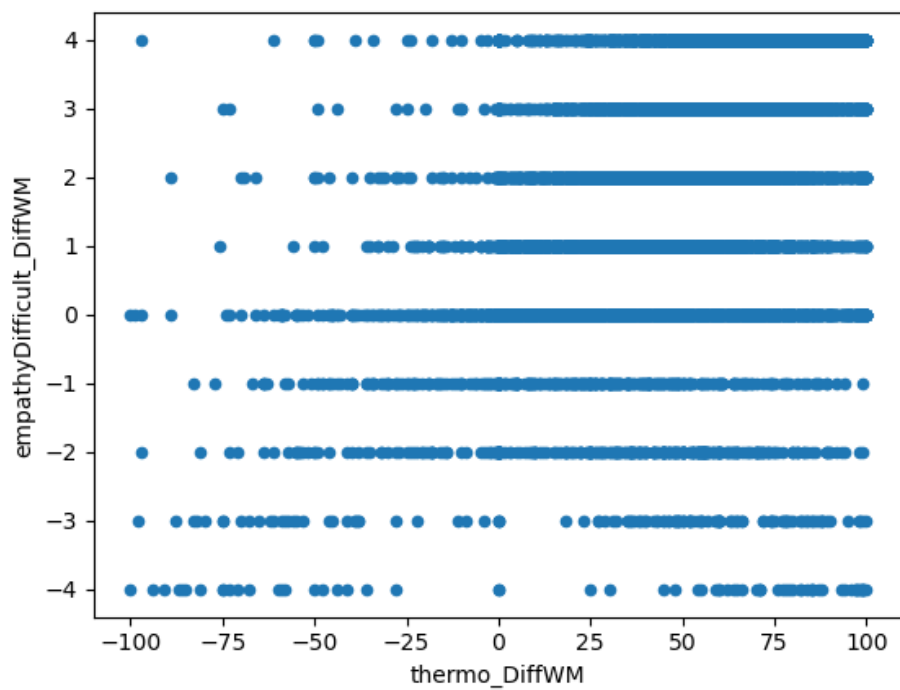


Figure 10: Thermometer Difference and Empathy Difference

4 Conclusion

Overall, I was able to successfully replicate the main results from the original paper, finding a statistically significant effect on affective political polarization from subscribing Facebook users to counter-attitudinal news in their feed. Tree-based models gave hints at a potential heterogeneous treatment effect among the treated, since baseline feelings thermometer and empathy difficulties differences between own-party and other-party were the most significant determinants of affective polarization. Specifically, the RCT design means a minimal set of assumptions needed for the identification of a causal effect. The coefficient values expressed in OLS are therefore unbiased estimates of the treatment effect of counter and pro-attitudinal interventions on affective polarization. Applications of causal machine learning models to estimate the average treatment effect gave very similar estimates to the original ATE, indicating the robustness of the statistically significant but small average treatment effect. However, estimates of the conditional average treatment effect by prior measures of polarization were inconclusive, indicating both the complexity of the underlying problem and the need for further development and standardization of causal machine learning techniques.

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